

100 Days of Linux - Day 037

Title: Editing Text with sed

Today's Goal

Learn how to search, replace, and edit text using the **sed** (Stream Editor) command.

Why This Matters

Developers and data engineers use sed to update configuration files, clean datasets, and automate text replacements without opening an editor.

Command of the Day: sed

Syntax: sed 's/old/new/' filename

Common Examples:

```
sed 's/Apple/Mango/' fruits.txt (replace first match per line)
sed 's/Apple/Mango/g' fruits.txt (replace all matches)
sed -n '1,5p' employees.txt (print lines 1-5)
sed -i 's/old/new/g' file.txt (edit file in place)
```

Explanation: sed reads text, applies editing rules, and prints the result.

Beginner Lesson

By default, sed displays the modified output without changing the original file. Use **-i** only when you want to save the changes directly to the file.

Practice Questions

1. Create fruits.txt with several occurrences of the word Apple. Replace the first Apple on each line with Mango using sed.
2. Replace every occurrence of Apple with Mango using the global option.
3. Create employees.txt with at least 10 lines. Display only lines 1 to 5 using sed.
4. Create config.txt containing the word development several times. Replace development with production directly in the file using sed -i, then verify the changes with cat.
5. Display your current working directory after completing today's tasks.

Hints

Remember: 's' means substitute, 'g' means replace all matches, and -i edits the file itself.

Mini Challenge

Create application.log with INFO, WARNING, and ERROR messages. Use sed to replace ERROR with CRITICAL, print only the first five lines, and verify the final output with cat.

Common Mistakes

Using **-i** before testing your command. First run **sed** without **-i** to preview the output.

Did You Know?

The name '**sed**' stands for **Stream Editor** because it edits text as it flows through a stream.

Pro Tip

Test your replacement command first, then add **-i** only after you're sure the output is correct.

Answer Key (Instructor Only)

Q1: **sed 's/Apple/Mango/' fruits.txt**

Q2: **sed 's/Apple/Mango/g' fruits.txt**

Q3: **sed -n '1,5p' employees.txt**

Q4: **sed -i 's/development/production/g' config.txt && cat config.txt**

Q5: **pwd**

Homework

Create three text files and practice replacing words, printing line ranges, and safely editing files with **sed**.

Next Day Preview

Tomorrow you'll learn **awk** to process structured text and extract columns from data files.